PHYS 570: Fusion Categories for Physics

▾ Course Description

An introduction to fusion categories and their applications to physics. Students will learn to work with fusion categories and their algebraic precursors with an emphasis on graphical and computational methods with the goal of preparing for research in theoretical/mathematical physics. Applications to be covered may include but are not limited to: anyon models and topological phases of matter in (2+1)D, topological quantum field theory, generalized symmetries in (1+1)D, quantum error correction, topological quantum computation, superselection theory, and (1+1)D conformal field theory.

▾ Syllabus

Click here to download the syllabus.

Course Schedule

The pace of the course will be based on the participants and the schedule will be adjusted accordingly. Suggestions for further reading to reinforce each week's lecture content will be posted as the course progresses.

	Tuesday	**Thursday**	**Supporting Reading**
Week 1	8/26. Course overview; Groups, Invertible Symmetries, and Categories (Part I)	8/28. Groups, Invertible Symmetries, and Categories (Part II)	I.1-I.2 in Zee's Group Theory for Physicists Chapters 1-3 in Abstract Algebra, Dummit and Foote
Week 2	9/2. Groups, Invertible Symmetries, and Categories (Part III)	9/4. The $\mathbb{Z}_2$ toric code	
Week 3	9/9. Representation theory, Noninvertible Symmetries, and Fusion Rings (Part I)	9/11. Representation theory, Noninvertible Symmetries, and Fusion Rings (Part II) Unit 1 Exercises Due	
Week 4	9/9. Representation theory, Noninvertible Symmetries, and Fusion Rings (Part III)	9/11. Kitaev Quantum Double Model	

Week 5	9/16. Fusion Categories, TQFTs, and Anyons (Part I)	9/18. Fusion Categories, TQFTs, and Anyons (Part II) Unit 2 Exercises Due	
Week 6	9/23. Fusion Categories, TQFTs, and Anyons (Part III)	9/25. Levin-Wen Models	
Week 7	9/30. Turaev-Viro-Barrett-Westbury state-sum TQFT	10/2. Fusion Categories, TQFTs, and Anyons (Part IV) Unit 3 Exercises Due	
Week 8	10/7. No class (Fall Break)	10/9. Fusion Categories, TQFTs, and Anyons (Part V)	
Week 9	10/14. Topological Quantum Computation (Part I)	10/16. Topological Quantum Computation (Part II) Unit 4 Exercises Due	
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Week 15	12/2. Advanced Topics Presentations	12/4. Advanced Topics Presentations	
Week 16	12/9. Advanced Topics Presentations	12/11. Advanced Topics Presentations	